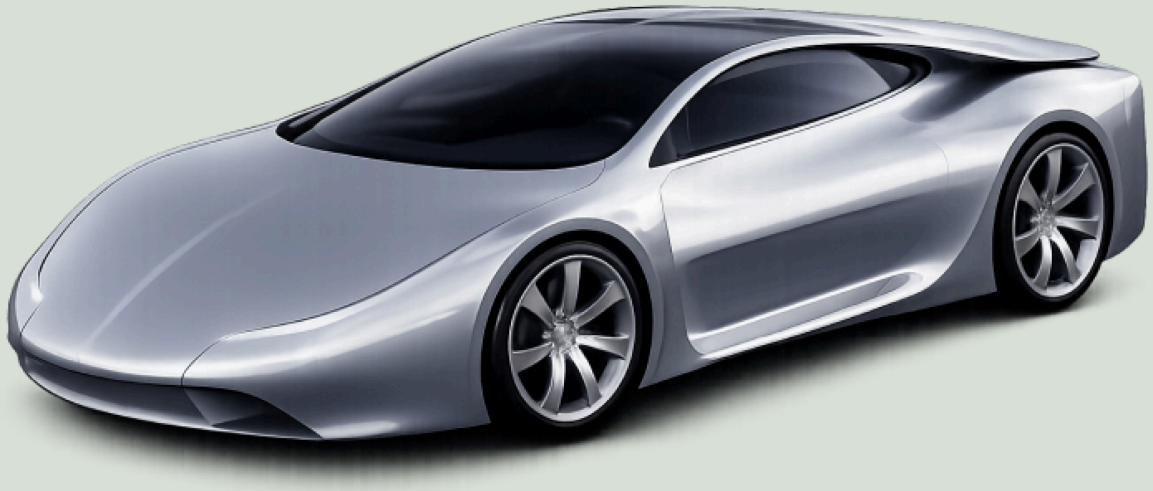
Ideas into physical form.





DESIGN ENGINEERING

A selection of products, spaces and engineering studies.



MECHANICAL DESIGN / ANALYSIS / PROTOTYPING

FIVE PROJECTS · 2026

Curiosity, backed by method.



Simon Xia · Design Engineering undergraduate

I work between the shape of an idea and the evidence that makes it credible.

I'm Simon Xia, a Design Engineering undergraduate at Imperial College London. I combine **geometric modelling, engineering analysis and physical prototyping** to investigate design decisions and communicate how products work.

PLACEMENT FOCUS

Mechanical engineering and engineering analysis: CAD development, simulation, prototyping and design evaluation.

EDUCATION

Imperial College London
Design Engineering · Four-year degree
2024–2028 (expected)

EDUCATION

Guanghua International School
Shanghai, China

SX1224@IC.AC.UK / +44 7876 529561

A practical toolkit.

To model, question, test and communicate.

01 **CAD & visualisation**

Fusion 360 · Blender

Parametric models, assemblies, surfaces and product rendering.

02 **Engineering analysis**

ANSYS · MATLAB

CFD, numerical calculations and interpretation of results.

03 **Programming**

Python · C++ · JavaScript

Computational problem-solving, scripting and interactive systems.

04 **Electronics & interfaces**

KiCad · Figma

PCB design, interface layouts and digital prototypes.

SELECTED WORK

01 STREAM X — Surface modelling & CFD — 04–07

02 AxisPro — Electromechanical product development — 08–11

03 ModuFrame — Circular aircraft-seat concept — 12–15

04 Jìng — The Path — Co-design & physical prototyping — 16–18

05 SIMREST Q2 — Product form & interface studies — 19–21

STREAM X

Shaping a vehicle around airflow.

A vehicle-design study exploring a smooth body and an integrated roof channel. Sketch development, surface modelling and a simplified CFD study connect the visual concept to pressure distribution, velocity and rear-wake behaviour.

ROLE

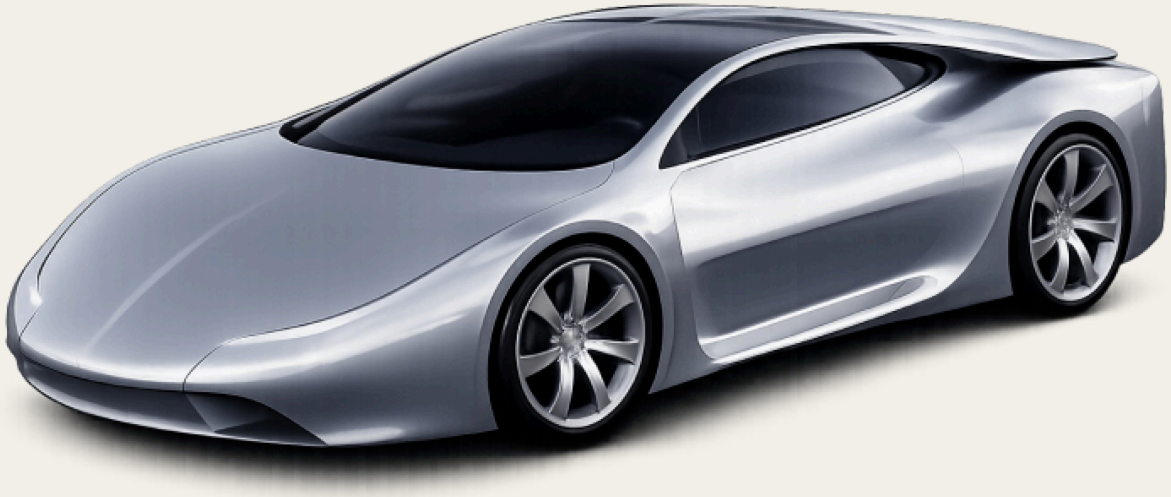
Individual design study by Simon Xia.

CONTEXT

DE2 Thermofluids: Energy and Design.

FOCUS

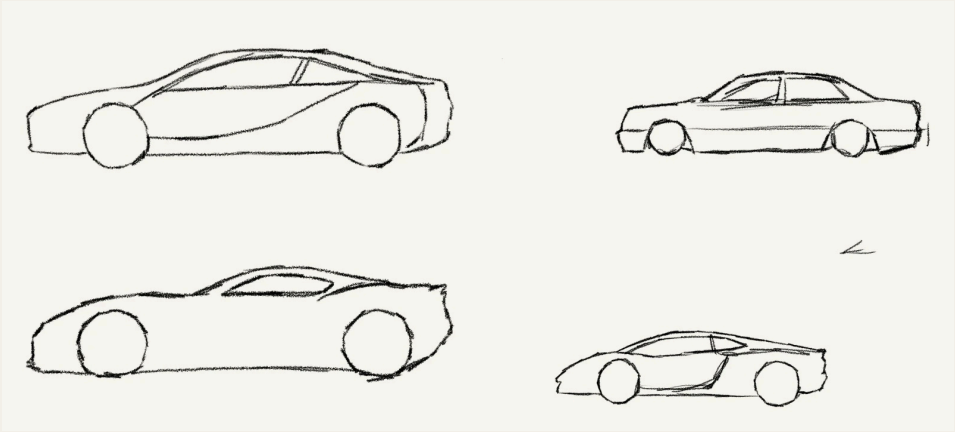
Form development · CAD · ANSYS Fluent analysis.



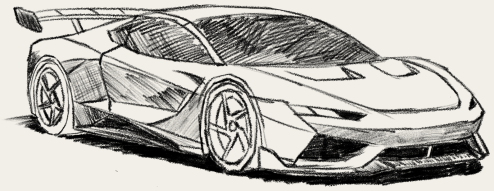
Vizcom AI-assisted presentation render. Original CAD and CFD evidence follows.

The form follows a question.

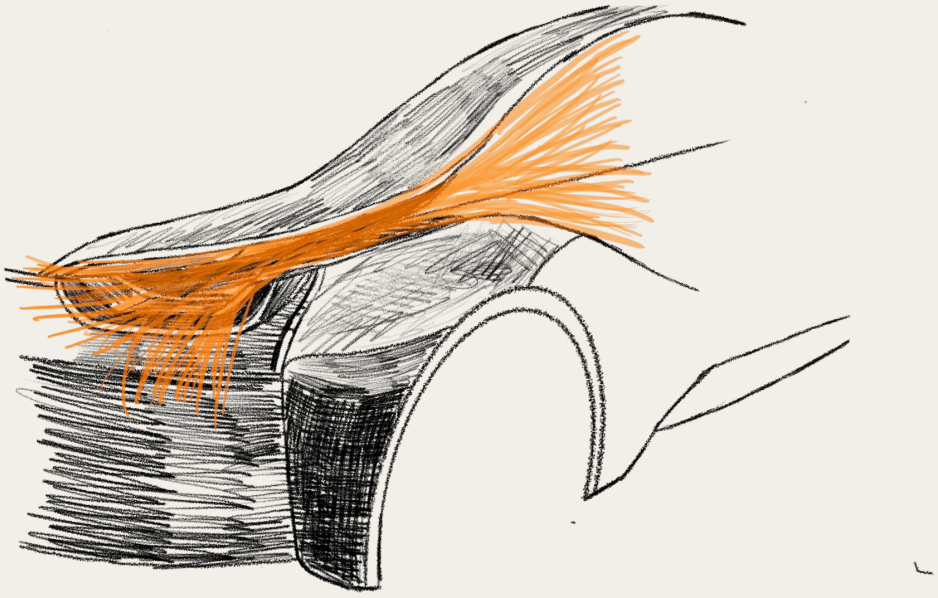
Early studies compared proportions and aerodynamic devices. An iteration with an external rear wing led to a different approach: **a C-shaped channel integrated into the roof and rear body**. The objective was to preserve a continuous surface while defining a route for the airflow.



01 · Initial side-profile studies.

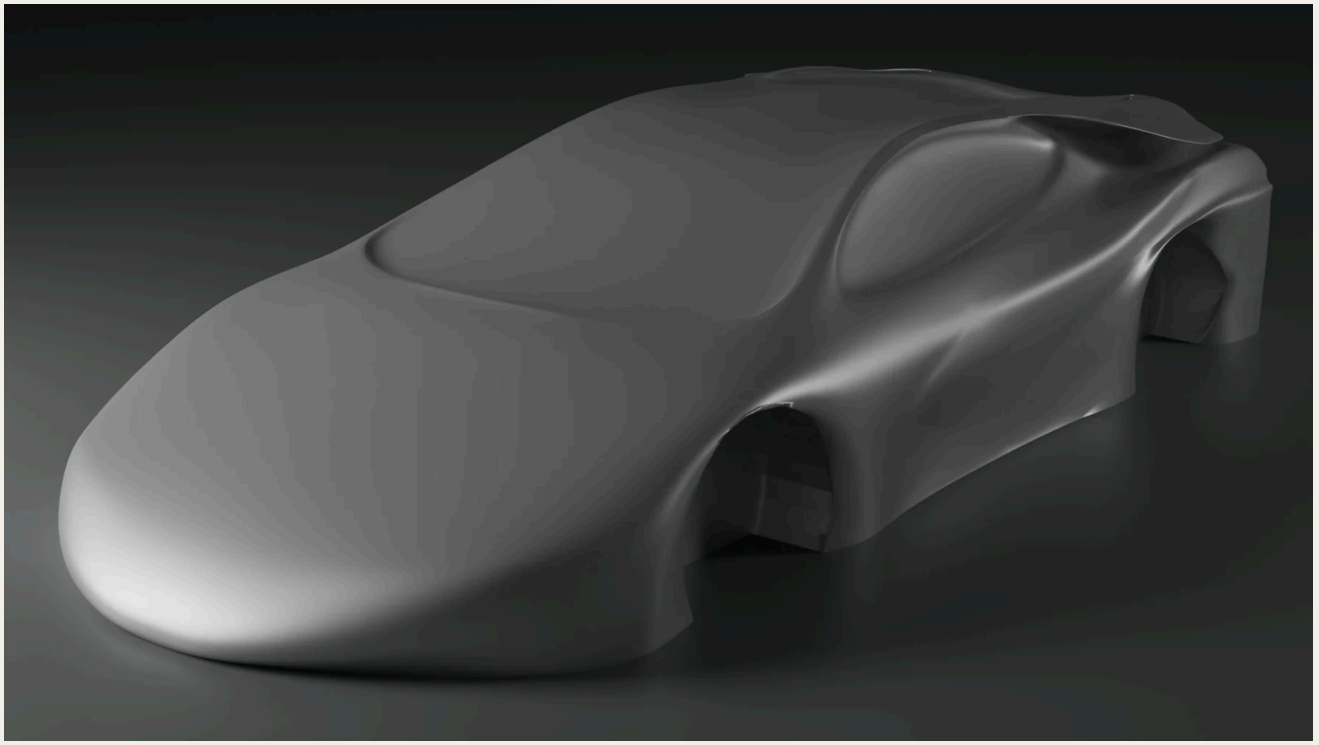


02 / Test an external wing.

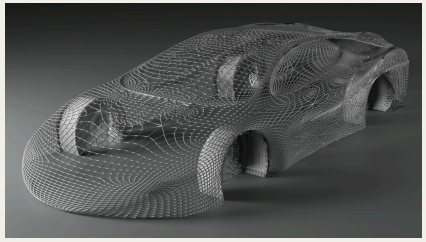


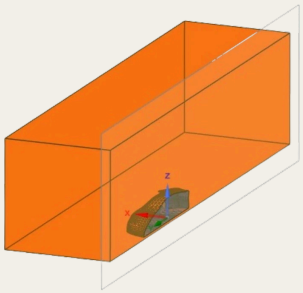
03 / Integrate the airflow route into the roof.

From surface to simulation.



Body-surface model in a clay material.





Geometry first.
Then a testable model.

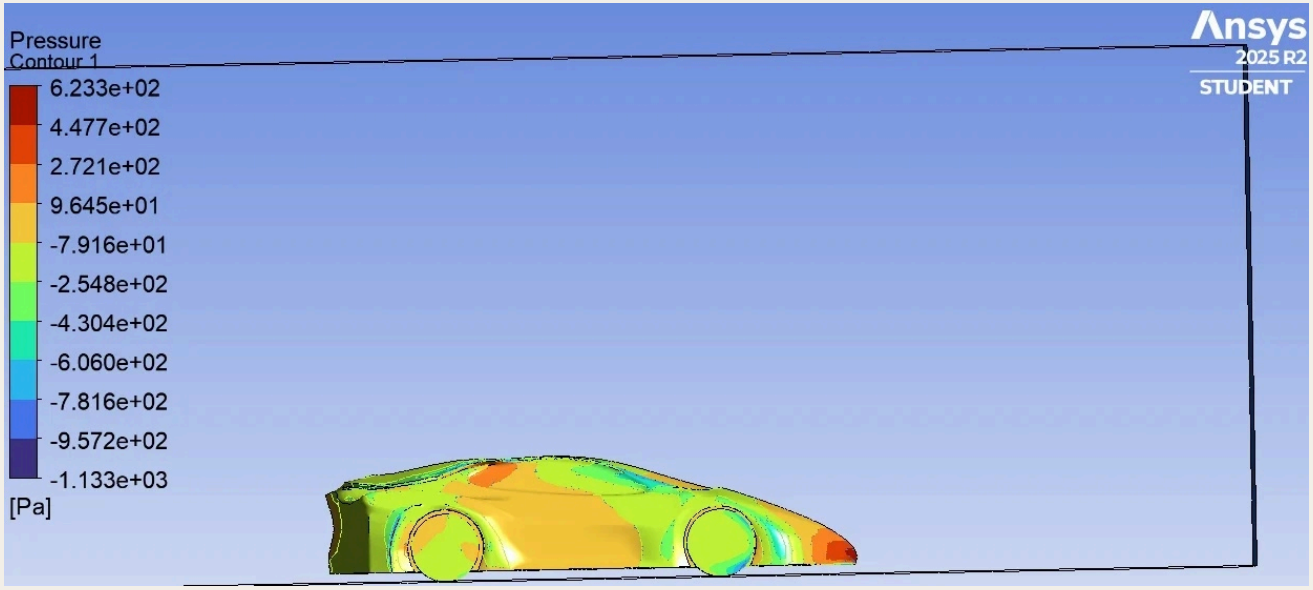
The clay and wireframe studies develop a low nose, curved cabin, wheel openings and rear channel. A **simplified half-model** used symmetry to reduce the CFD domain. Pressure and velocity fields were then reviewed in **ANSYS Fluent**.

ANALYSIS SETUP

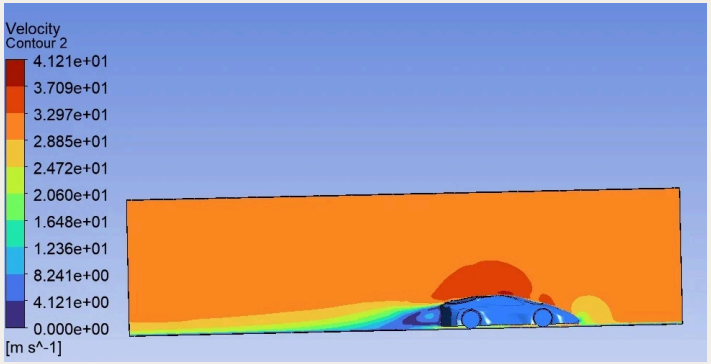
30.5 m/s inlet
k–ε turbulence model
458,336-cell mesh
Symmetry-based half-model

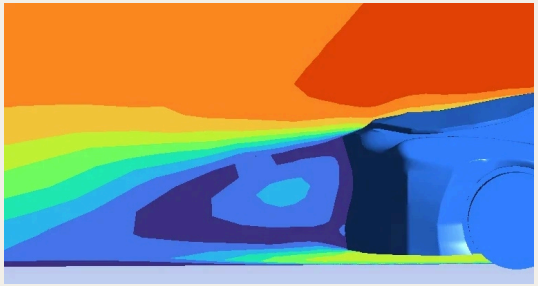
Reduced-scale study with a stationary, no-slip road.

Flow fields become design decisions.



Pressure contour from the original report; original legend retained.





SIMULATED DRAG COEFFICIENT

0.345

Simplified study model

Leading-edge stagnation, channel flow and rear-wake separation informed the next iteration.

NEXT ITERATION

Refine the channel exit, underbody and rear lip; compare under consistent simulation conditions.

Indicative reduced-scale CFD result. Full-scale performance has not been validated.

AxisPro

A motorised workstation for miniature painting.

Developed over four months, AxisPro combines height adjustment, tilt, controlled rotation and a secure model clamp. Team research, physical prototypes and ergonomic trials shaped a compact workstation with direct physical controls.

MY CONTRIBUTION

Led **business-model development, product labelling, the user guide and rendering**; contributed to the wider team design process.

CONTEXT

Four-person team · Product Design Engineering · Imperial College London.



Product visualisation · Team: Aaron Xu, Aidan Ryder, Lucian Brand and Simon Xia.

Start with the painter.

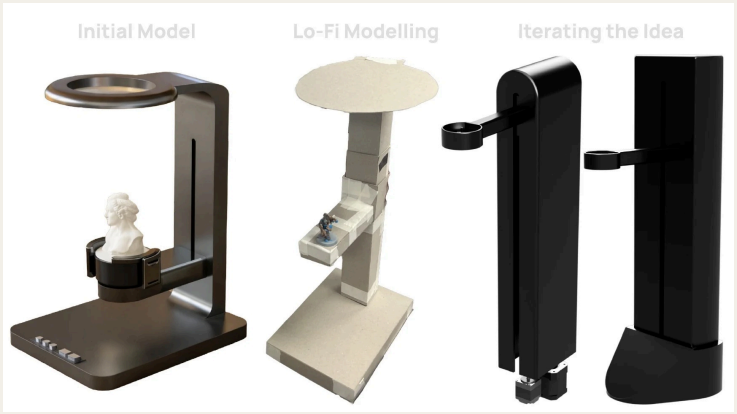
The team investigated the workpiece, hand position, posture and workspace. **Greyboard models tested comfortable painting heights** before the mechanism was committed to CAD. Telescoping, belt, pulley and lead-screw approaches were compared; a lead screw offered a simpler, self-locking direction.



Adjustable greyboard model in a painting-height trial.



The same mock-up tested with a taller participant.



Concept, physical mock-up and successive tower studies.

01 / OBSERVE

Understand posture, reach, painting height and the working footprint.

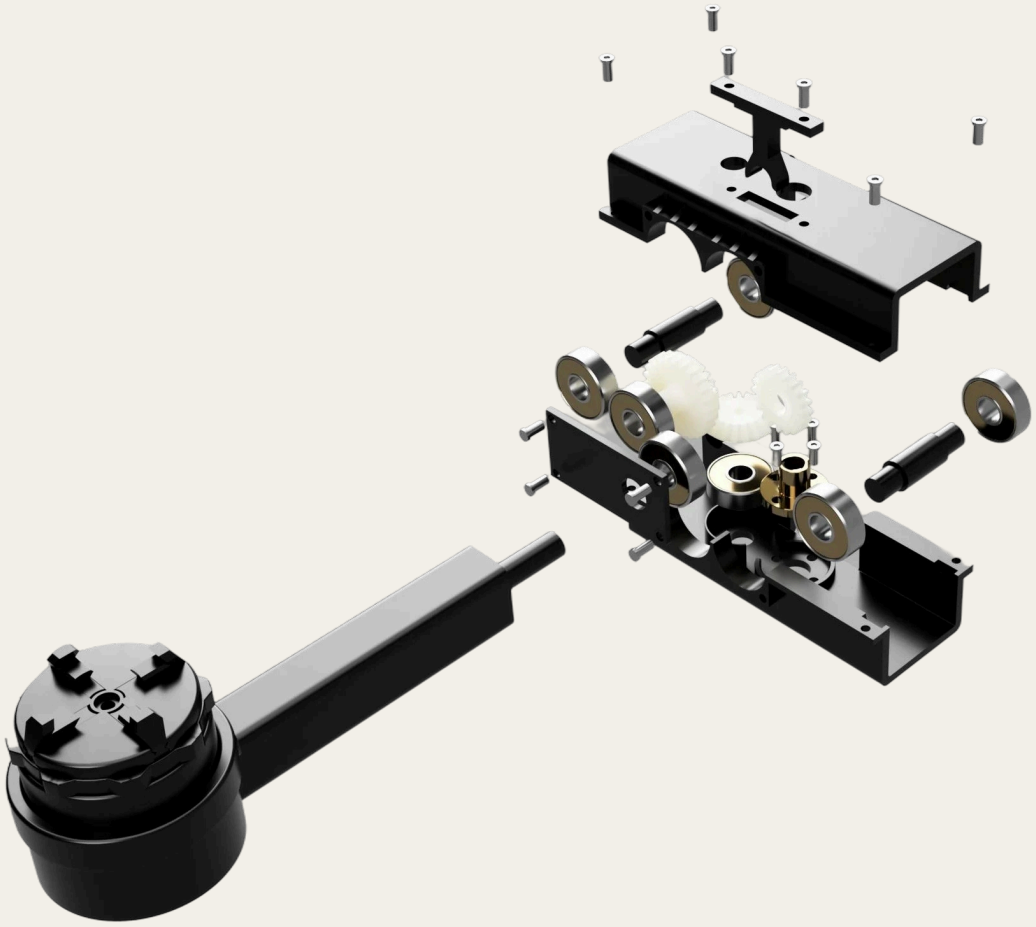
02 / PROTOTYPE

Use greyboard mock-ups to test the working position before detailed CAD.

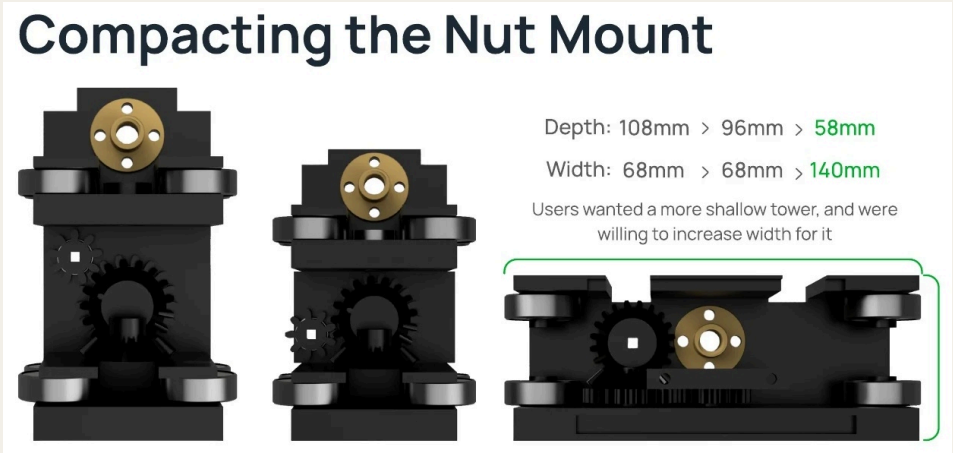
03 / SELECT

Compare mechanisms and develop a simpler, self-locking lead-screw direction.

Refine the system, one interface at a time.



Exploded CAD study of the holder, bearings, gears and mount.



Nut-mount iterations responding to footprint feedback.

MOUNT DEPTH

108 → 58 mm

A wider nut mount reduced tower depth **in response to user feedback**. Control trials informed the joystick and rotary-encoder arrangement.

Team development · Ergonomic feedback guided mechanism packaging and repair access.

Make the product understandable.



The assembled prototype in a miniature-painting workflow.

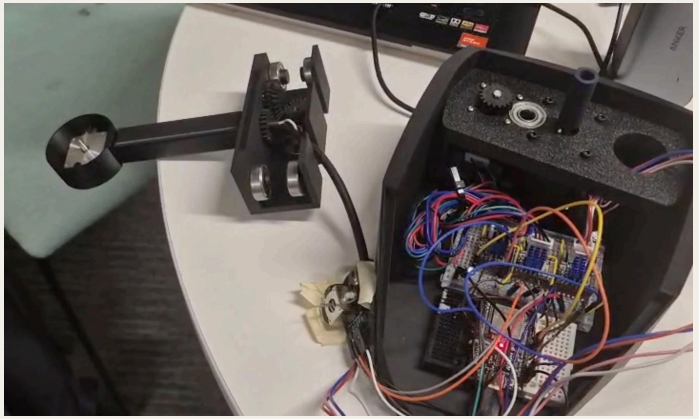
The assembled prototype combines height, tilt and rotation with a secure clamp. Further work includes moving-wire management and continued ergonomic evaluation.

MY CONTRIBUTION

Rendering, product labels, a **five-page user guide** and business-model development.

WHAT THE GUIDE RESOLVES

Setup, operation, safety and maintenance: helping a complex mechanism become a usable product.



Working prototype and electronics integration are team outcomes.

ModuFrame

An aircraft seat designed for component-level renewal.

An economy-class aircraft-seat concept separating an aluminium frame from replaceable comfort layers and accessories. System mapping, modular architecture, material choices and circularity assessment guide the design.

ROLE

Individual design development by Simon Xia, informed by group research.

CONTEXT

DE2 Sustainable Design Engineering · Part 2 individual submission.



AI-assisted concept renders based on the original design studies.

Keep the structure. Renew the rest.

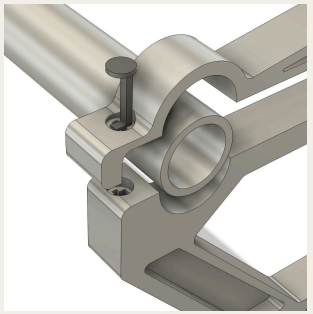


AI-assisted concept render · Three-seat primary frame.

Separation is a design requirement.

An aluminium 6061 frame combines X-shaped legs, curved back supports and modular seat-pan connections.

Proposed bolted and quarter-turn interfaces separate the structural frame from higher-turnover components.





Reversible joint and detachable seat-pan concepts.

Comfort as replaceable layers.

The cushion, backrest, headrest and armrest contact surfaces are separated into **removable components**. Proposed rPET, TPU and E-TPU layers support cleaning and replacement. The tray, power unit and storage are treated as independent accessories so the architecture can accommodate maintenance and upgrades.



01 / Cushion layers · AI-assisted concept render.



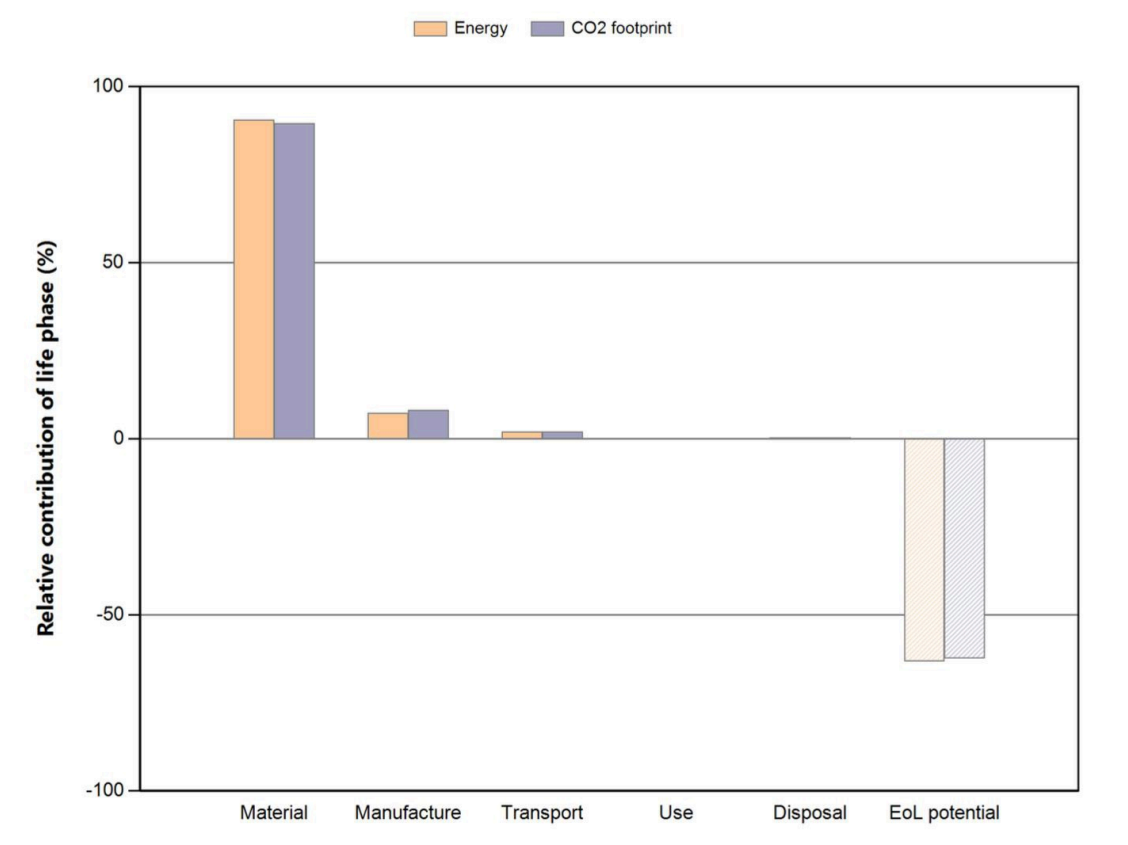
02 / Backrest layers.



03 / Service modules.

Next design decision · Replace the storage bag’s adhesive attachment with a **reversible interface.**

Material choices, made explicit.



Source Eco Audit: materials, manufacture, transport and end-of-life scenarios.

REPORTED WEIGHTED MCI

0.70

Aluminium dominates the reported mass and embodied impact. The assessment directs further work towards lower-carbon material choices, reversible joints and material tracking.

METHOD

Mass-weighted Material Circularity Indicator and Ansys Granta EduPack Eco Audit.

Scope · Concept assessment; source Eco Audit assumes one year and omits operational use. Testing and certification remain future work.

Jìng — The Path

A more inviting place to pause on Queen’s Lawn.

The team explored how campus space could support study, rest and conversation beyond sunny days. Observation, interviews and concept comparison led to a lower path combining curved shelter, modular seating and open circulation.

ROLE & CONTEXT

Team member · Five-person design study · Queen’s Lawn, Imperial College London.

METHODS

Observation · Interviews · **Co-design** · CAD · 3D printing · **Full-scale cardboard prototyping**.



Campus proposal · Team: Simon Xia, Yuxuan Zhou, Cici Song, Ziqi Zhou and Simon Wang.

A conversation through prototypes.

A 3D-printed seating model explored modular arrangements; a full-size cardboard version allowed informal sitting and reclining feedback. Foam and cardboard site models then compared a raised loop with a lower path. Twelve participants — eleven students and one professor — contributed to the co-design session.



Handling the 3D-printed seating model.



Full-scale cardboard comfort trial.



Elevated-loop model.



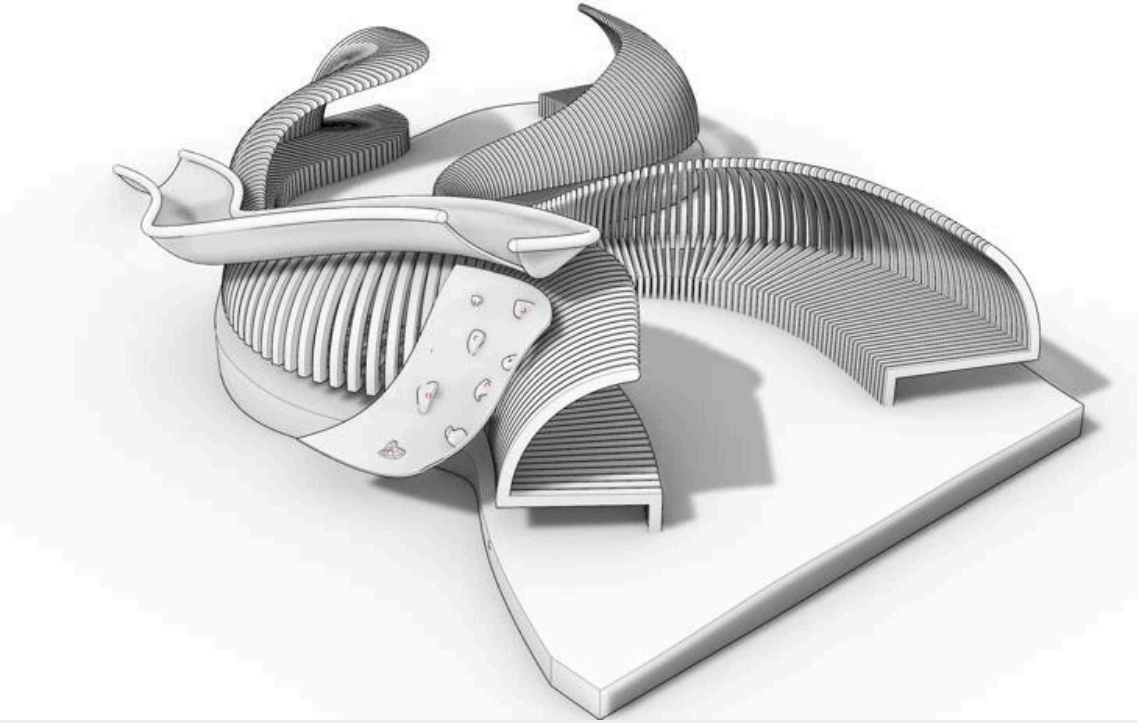
Lower-path model.

8 / 12

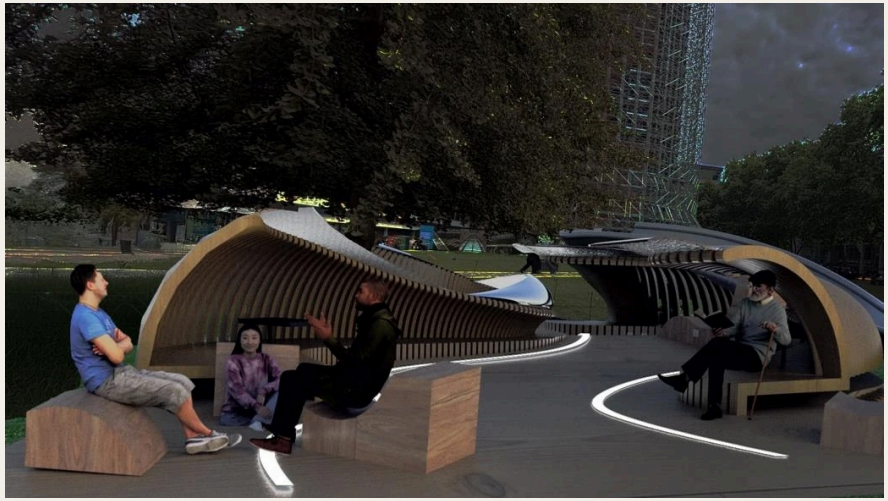
participants preferred the lower path.

Informal co-design feedback informed the next spatial iteration.

Connect places to pause.



Detachable seating, quieter and shared zones, and **proposed drainage** turn the initial concept into a more considered campus landscape.



Design proposal. Scale and seating prototypes were made; the full installation has not been built.

REFINEMENT

Balance shelter, circulation and playful routes with spaces to study, rest and talk.

SIMREST Q2

A quiet object for the bedside.

A bedside-device concept combining a charging surface, light feedback and physical controls. The study explores how a calm exterior can accommodate tactile interaction, connection points and an understandable component arrangement.

ROLE

Individual concept design by Simon Xia.

FOCUS

Product form · Physical interfaces · Assembly visualisation.

SCOPE

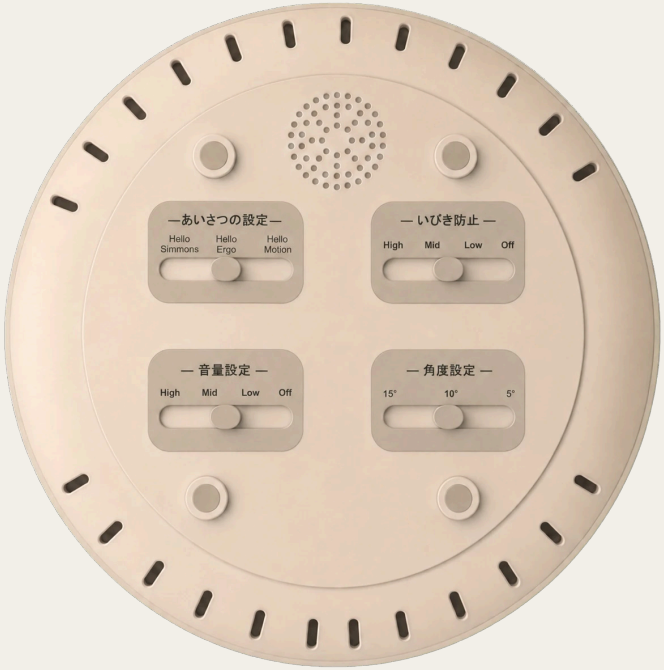
Concept renders and design studies; no physical or clinical performance claim.



Concept visualisation · Product form and physical interface study.

A calm surface. A legible interface.

Detailed settings move to the underside. Two alternatives compare tactile position with illuminated mode labels, keeping the visible top surface clear.



A / SLIDERS

Position communicates the selected setting.



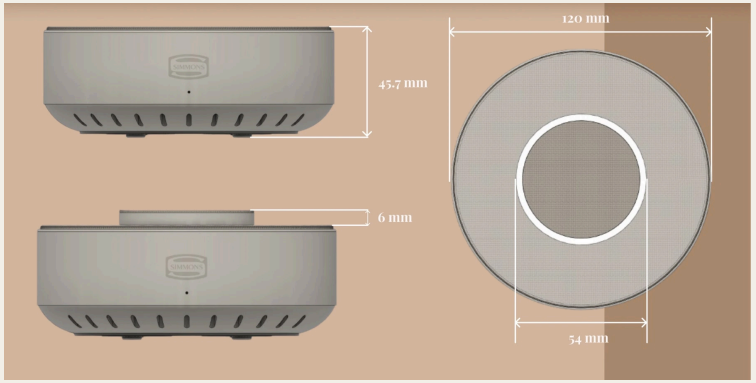
B / BUTTONS

An **illuminated label** communicates the active mode.

Resolve the object from the inside out.



Exploded concept assembly: surface, light ring, textile, shell and base.



PROPORTIONS

120 mm diameter
45.7 mm body height

DESIGN INTENT

A compact enclosure with layered functions, understandable controls and a coherent assembly.

Proposed geometry and component arrangement; physical validation remains future work.

Thoughtful design.
Grounded in engineering.

I’m looking to contribute through CAD development, engineering analysis, prototyping and clear design communication.

sx1224@ic.ac.uk

Explore the interactive portfolio

+44 7876 529561

Imperial College London
Design Engineering · 2024–2028 (expected)

Image credits: STREAM X presentation renders · Vizcom AI. MODUFRAME component renders · AI-assisted concept visualisation.